

Supply Chain Project - Power BI

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Project Overview

This project analyzes order fulfillment, shipping performance, and profitability across **65,752 unique orders** (180,519 total line items). The goal is to uncover why shipping delays happen, identify fulfillment bottlenecks, and optimize operational performance.

Key Skills Used

- **Power BI & Data Visualization:** Interactive dashboard design, custom layout, and visual formatting.
- **DAX:** Custom calculated measures for Late Delivery %, Profit Margin %, Total Sales, and Total Profit.
- **Data Cleaning & Preprocessing:** Filtering errors, removing blank rows/unnecessary columns, correcting data types, and retaining loss-making negative profit values.

Key Performance Indicators (KPIs)

- **Total Sales: \$36.78M**
- **Total Profit: \$3.97M**
- **Profit Margin: 10.78%** (*Net Margin: 12.00%*)
- **Total Orders: 66K** (*65,752 unique orders*)
- **Total Late Orders: 36K**
- **Overall Late Delivery Rate: 54.82%**

Core Problem Analysis & Business Insights

1. Why Are Deliveries Delayed?

- **First Class Has the Highest Delays: First Class shipping** is the biggest cause of delivery delays, arriving late **95.3%** of the time.
- **Unrealistic Delivery Targets:** The main reason for delays is that promised delivery dates are set too fast for actual shipping speeds:

- **First Class:** Promised in **1 day**, but actual shipping takes **2 days** (arriving late **95.3%** of the time).
- **Second Class:** Promised in **2 days**, but actual shipping takes **4 days** (arriving late **76.6%** of the time).
- **Same Day:** Promised in **0 days**, but actual shipping takes about half a day (**0.48 days**), leading to a **45.7%** late rate.
- **Standard Class Works Best:** **Standard Class** allows **4 days** and actually takes **3.99 days**, making it the most reliable option with the lowest late rate (**38.1%**).

2. Where Are the Bottlenecks?

- **Expedited Shipping Tiers:** **First Class** and **Second Class** fulfillment fail to meet promised delivery windows over **75% of the time**.
- **High-Risk Regions:** Delays concentrate heavily in **Central Africa (58.0%)**, **South Asia (56.3%)**, **East Africa (55.9%)**, and **Western Europe (55.8%)**.

3. How to Improve Performance & Profitability

- **Re-align Delivery Targets:** Set First Class delivery expectations to **2 days** and Second Class to **4 days** at checkout to match real shipping speeds.
- **Target Regional Logistics:** Renegotiate local carrier contracts and fulfillment partnerships in high-delay regions (**Central Africa** and **South Asia**).
- **Prioritize High-Margin Orders:** Fast-track warehouse dispatch for top profit-generating items (to prevent costly order cancellations).

Data Cleaning & Transformation

1. Removed blank rows, data errors, and irrelevant columns.
2. Formatted numerical, date, and text fields correctly.
3. Retained negative profit entries to capture real business losses.
4. Created calculated columns and DAX measures for delay tracking and profit margin analysis.